

Continual Learning and Model Editing for Foundation Models: A Comprehensive Survey

Abstract

The emergence of large language models and multimodal foundation models has fundamentally altered the paradigm of artificial intelligence. By leveraging massive-scale pre-training on diverse corpora, these architectures exhibit extraordinary emergent capabilities, reasoning proficiency, and cross-domain generalization. However, the static nature of the pre-training paradigm imposes a critical bottleneck: once a model is deployed, its parametric knowledge is frozen. In a dynamically evolving world characterized by continuous factual updates, shifting data distributions, and changing human preferences, deploying static models inevitably leads to knowledge staleness, factual hallucinations, and deteriorating performance. Retraining foundation models from scratch to accommodate new data is computationally prohibitive and environmentally unsustainable. Consequently, post-deployment adaptation techniques have become paramount. This manuscript presents a comprehensive academic survey of two primary paradigms addressing this challenge: Continual Learning and Model Editing. Continual learning approaches aim to facilitate the sequential acquisition of new tasks or domains over time while mitigating catastrophic forgetting, addressing the classic stability-plasticity dilemma across the entire model lifecycle—from continual pre-training to instruction tuning and preference alignment. Conversely, model editing seeks to apply precise, post-hoc interventions to alter specific factual associations within the model’s weights without disrupting unrelated global behaviors. This survey systematically categorizes the methodological families within both disciplines, ranging from parameter-efficient adapters, dynamic prompting, and mixture-of-experts architectures to locate-and-edit algorithms, hypernetworks, and in-context retrieval frameworks. Furthermore, the analysis extends to highly complex, contemporary frontiers, including multimodal knowledge editing, cross-lingual propagation, multi-hop ripple effects, and the critical intersection of model editing with machine unlearning. By synthesizing empirical findings and mechanistic insights, this survey provides a structural foundation for advancing the development of autonomous, self-evolving artificial intelligence systems capable of lifelong adaptation.

Introduction

Foundation models, including large language models (LLMs) and vision-language models (VLMs), have demonstrated unprecedented success across a broad spectrum of cognitive and generative tasks¹. Trained on internet-scale datasets via self-supervised objectives, these architectures encapsulate vast amounts of linguistic patterns, factual knowledge, and reasoning heuristics. Despite these achievements, a fundamental limitation persists: the

parametric knowledge of a foundation model reflects only the state of the world at the time of its training⁴. As new events occur, scientific consensus evolves, and human values shift, the pre-trained knowledge base inevitably becomes outdated. This temporal misalignment leads to performance degradation, factual hallucinations, and the propagation of obsolete or erroneous information⁵.

Addressing knowledge staleness through complete model retraining is economically and computationally intractable due to the massive scale of contemporary architectures, which often feature hundreds of billions of parameters². To bridge the gap between static pre-training and dynamic real-world environments, the artificial intelligence research community has increasingly focused on computationally efficient post-training adaptation frameworks. The two most prominent paradigms in this domain are Continual Learning (CL) and Model Editing (ME)².

Continual learning, historically referred to as lifelong or incremental learning, is a machine learning paradigm that enables artificial neural networks to learn sequentially from continuous data streams². The central objective of continual learning is to integrate new knowledge, domains, or skills while preserving previously acquired capabilities. This endeavor is fundamentally constrained by the "stability-plasticity dilemma." An excess of plasticity causes the model to overwrite existing parameters, resulting in "catastrophic forgetting"—the rapid and severe degradation of historical knowledge. Conversely, excessive stability prevents the model from effectively adapting to novel data². Modern continual learning for foundation models spans multiple stages of the model lifecycle, including continual pre-training, continual instruction tuning, and continual alignment².

In contrast, model editing (also known as knowledge editing) addresses model adaptation at the instance level. Rather than sequentially training the model on extensive new datasets, model editing provides targeted, surgical interventions to update or correct specific factual associations¹⁵. The objective is to efficiently alter a designated fact (e.g., updating the CEO of a corporation) while rigorously preserving the model's performance on unrelated queries. Model editing algorithms range from direct parameter modifications and hypernetwork-driven gradient transformations to non-parametric, retrieval-augmented interventions⁷.

While both disciplines seek to adapt foundation models dynamically without full retraining, they operate at different granularities and encounter distinct methodological hurdles. Furthermore, these domains increasingly intersect with machine unlearning, which aims to selectively eradicate hazardous, copyrighted, or sensitive information from the latent space of the model¹⁸. A synthesized understanding of continual learning and model editing is essential for building trustworthy, self-evolving foundation models.

This survey systematically reviews the state-of-the-art in continual learning and model editing. The subsequent sections are organized as follows: The Methods section delineates the core stages of continual learning (pre-training, instruction tuning, alignment) and categorizes the predominant algorithmic families in both continual learning and model editing. It further explores extensions into multimodal and cross-lingual domains. The Challenges and Prospects section critically examines open problems, including multi-hop ripple effects, sequential degradation, and the nuanced distinction between model editing and machine unlearning.

Finally, the Conclusion synthesizes the findings and outlines future trajectories for lifelong learning in artificial intelligence.

Methods

The landscape of post-deployment model adaptation can be systematically organized into distinct methodological families. This section first explores the multi-stage pipeline of continual learning in foundation models, followed by a taxonomy of parameter-efficient and architectural continual learning algorithms. It then transitions to the predominant methodologies employed in model editing, culminating in an analysis of multimodal and cross-lingual extensions.

The Stages of Continual Learning in Foundation Models

Unlike classical incremental learning, which typically focuses on narrow classification tasks, continual learning for foundation models encompasses a hybrid, multi-stage training pipeline. This pipeline dictates how a model assimilates new domain corpora, acquires new instructional behaviors, and aligns with shifting human preferences².

Continual Pre-Training (CPT)

Continual Pre-Training (CPT) involves updating an existing, general-purpose foundation model using large-scale, domain-specific corpora (e.g., biomedical, financial, or legal texts) to create specialized models without the immense cost of training from scratch⁵. Standard pre-training assumes an independent and identically distributed (i.i.d.) dataset, whereas CPT introduces severe sequential distribution shifts.

A primary obstacle in CPT is the "stability gap," a phenomenon characterized by a chaotic spike in the training loss immediately following the introduction of a new data distribution²².

Mechanistic analyses reveal that this instability is highly sensitive to learning rate scheduling. Research conducted on models such as Pythia and LLaMA demonstrates that progressively re-warming the learning rate, rather than allowing it to decay to zero, is vital for computational efficiency when resuming training on novel datasets²². However, abruptly applying the maximum learning rate exacerbates the stability gap, causing the model to rapidly overwrite its foundational representations. To mitigate this, Context-Aware Continual Pre-Training (CA-CPT) techniques have been developed. CA-CPT identifies that the initial tokens of a sequence generate disproportionately high losses during training, producing massive gradient variance. By masking these high-loss initial tokens during the learning process, the model stabilizes, effectively absorbing new domain knowledge while exhibiting a significantly lower forgetting rate on general benchmarks²⁵.

Data curation and mixture strategies further dictate the success of CPT. Continually pre-training an LLM on an entire massive domain corpus for a single epoch induces a prolonged plasticity gradient that overwrites prior weights²³. Empirical studies suggest that pre-training on high-quality subsets across multiple epochs accelerates performance recovery and preserves general capabilities. Furthermore, the integration of synthetically generated, multidisciplinary question-and-answer pairs into the CPT mixture has proven highly effective. For instance, incorporating synthetic scientific and code-based QA data during the continual pre-training of LLaMA-3 substantially improved scientific reasoning and mathematical

capabilities while preserving English linguistic proficiency¹⁴.

Continual Instruction Tuning (CIT)

Following pre-training, foundation models undergo instruction tuning to align their generative processes with human intent and task-specific constraints. Continual Instruction Tuning (CIT) enables LLMs to adapt incrementally to new tasks or APIs. However, sequential fine-tuning frequently triggers catastrophic forgetting, where the model loses its general instruction-following abilities or reasoning skills²⁷.

Recent studies offer a paradigm-shifting perspective on this phenomenon, distinguishing between true knowledge loss and "spurious forgetting"²⁹. Spurious forgetting posits that the performance degradation observed during sequential tasks often reflects a decline in *task alignment* rather than the actual erasure of underlying knowledge representations. During the initial optimization steps on a new instruction set, large gradients disrupt the shallow alignment between the model's deep semantic representations and its output layers. The underlying knowledge remains intact, but the model loses the formatting or alignment necessary to access it. This hypothesis is supported by findings that spurious forgetting can be rapidly reversed through minimal data replay or by employing a "Freeze" strategy, wherein the bottom layers of the transformer (approximately 30%) are frozen to protect core representations while allowing the output layers to adapt³⁰.

This mechanistic understanding is further formalized by the Instruction Vector (IV) framework. The IV framework maps fine-tuning dynamics by projecting instruction-following capabilities into specific geometric directions within the latent space³³. Analyses of these vectors reveal that fine-tuning generally introduces specialized reasoning pathways rather than destroying previous skill vectors. By implementing IV-guided training—which incorporates explicit regularization to preserve the original computational graph along older instruction vectors—models can mitigate catastrophic forgetting and maintain performance across diverse benchmarks³³.

Continual Alignment and Preference Learning

The final stage of LLM development involves aligning the model with ethical guidelines and human preferences, predominantly through Reinforcement Learning from Human Feedback (RLHF) or Direct Preference Optimization (DPO). Because human values, safety definitions, and user preferences are non-stationary, *continual alignment* is practically indispensable¹⁰.

Directly learning new preferences using traditional DPO algorithms on sequential, highly imbalanced preference data leads to severe catastrophic forgetting of historical alignment states³⁶. Gradient updates often favor the current majority signals, undermining prior safety guardrails and causing the model to generate helpful but potentially harmful outputs, or vice versa³⁷. To address these distributional biases, algorithms such as Continual Optimal Policy Regularization (COPR) have been introduced. COPR frames continual alignment as a constrained optimization problem, utilizing Lagrangian Duality to dynamically regularize the current policy against a historically optimal policy distribution. This prevents the model from deviating excessively toward new, unbalanced objectives and anchors the alignment state³⁵.

Concurrently, Fairness DPO (ϕ -DPO) explicitly integrates distributional bias penalties into the preference optimization loss, ensuring that sequential alignment remains robust and fair across diverse demographic and modality distributions without requiring complex reinforcement learning pipelines³⁶.

Methodological Families in Continual Learning

To engineer continuous adaptation without triggering catastrophic forgetting, researchers have developed various parameter-efficient, architecture-based, and regularization strategies specifically tailored to the extreme scale of foundation models. These approaches are summarized in Table 1 and detailed below.

Methodological Family	Core Mechanism	Representative Algorithms	Primary Advantages in Foundation Models
Prompt-Based Learning	Learning tunable soft prompts appended to inputs or self-attention keys while freezing the backbone.	L2P, DualPrompt, CODA-Prompt, VQ-Prompt	Completely rehearsal-free; minimal parameter overhead; leverages frozen, robust pre-trained representations.
Parameter-Efficient Adapters	Injecting sequential low-rank matrices or modular adapters into linear layers with subspace constraints.	ELLA, JumpLoRA, DEAL, PESO	Highly scalable; computationally efficient; enables explicit mathematical bounds on cross-task parameter interference.
Architectural / Mixture-of-Experts	Expanding capacity organically by routing inputs to task-specific or transient experts via gating networks.	LLaMA-MoE, CP-MoE, MoRAM, MoE-CL	High intrinsic capacity; mitigates weight interference structurally by isolating parameters at the routing level.

Regularization and Latent Replay	Penalizing drift mathematically or replaying highly compressed latent representations instead of raw data.	Latent Replay, TFGN	Bypasses raw data storage, alleviating privacy concerns; protects critical weights without exponential memory growth.
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Prompt-Based Continual Learning

Prompt-based continual learning leverages the robust, frozen representations of pre-trained models by training small sets of continuous parameters (prompts) that dynamically instruct the model conditioned on the input data⁴⁰. Because the underlying model weights are strictly preserved, these methods are highly resistant to catastrophic forgetting.

- **L2P (Learning to Prompt):** L2P establishes a shared prompt pool where the input feature acts as a query to retrieve the most relevant prompts. This key-query mechanism enables task-agnostic continual learning without requiring explicit task identifiers or a data rehearsal buffer during inference⁴².
- **DualPrompt:** Building upon L2P, DualPrompt defines a hierarchical prompting structure. It employs "general" prompts in the early transformer layers to capture task-invariant features and "expert" prompts in deeper layers to encode task-specific knowledge, thereby balancing stability and plasticity⁴⁴.
- **CODA-Prompt (Continual Decomposed Attention-based Prompting):** Addressing the plasticity limitations of static key-query mechanisms, CODA-Prompt decomposes prompts into a set of learnable components that are dynamically assembled via input-conditioned attention weights. This end-to-end optimization scheme provides greater expressivity and significantly enhances performance in rehearsal-free settings⁴⁶.
- **VQ-Prompt:** To stabilize task knowledge representation, Vector Quantization Prompting (VQ-Prompt) incorporates discrete quantization into the end-to-end training of prompts. By forcing continuous prompt generation into nearest-match discrete representations from a codebook, the method achieves superior abstraction and stability across sequential tasks⁴⁸.

Parameter-Efficient Fine-Tuning (PEFT) and Adapters

Low-Rank Adaptation (LoRA) has become the predominant technique for efficient LLM

fine-tuning by injecting trainable rank decomposition matrices ($\Delta W = AB$) while freezing the base model⁴⁹. In continual learning, however, standard sequential LoRA updates suffer from capacity saturation and severe cross-task gradient interference.

- **Subspace-Aware Isolation:** Methods such as ELLA (Efficient Lifelong Learning for Adapters) address interference by characterizing the structure of past updates. ELLA applies a lightweight regularizer that penalizes new updates from aligning with the

high-energy, task-specific directions of previous adapters. Unlike strict orthogonality methods that rapidly exhaust the available parameter space, ELLA preserves freedom in low-energy residual subspaces, bounding interference while enabling forward transfer of generic linguistic patterns⁵¹.

- **Dynamic Sparsity:** Frameworks like JumpLoRA leverage specialized activation functions, such as JumpReLU, to adaptively induce coordinate-wise sparsity within the LoRA blocks. By optimizing a threshold directly alongside the adapter parameters, JumpLoRA dynamically cancels redundant weights, ensuring different task adapters occupy disjoint parameter coordinates to prevent interference⁵³.

Mixture of Experts (MoE) Architectures

Mixture-of-Experts (MoE) architectures inherently support continual learning by routing diverse data distributions through specialized sub-networks (experts), thereby structurally isolating task parameters.

- **Consistency-Preserving MoE (CP-MoE):** Traditional LoRA-based MoE methods face a trade-off between isolating experts too aggressively (limiting forward transfer) or allowing overwriting. CP-MoE introduces a "transient expert" that captures high-variance, early task-specific updates. A routing bias then guides the integration of this knowledge into stable historical experts, significantly reducing parameter collision across task sequences⁵⁴.
- **MoRAM (Mixture of Rank-1 Associative Memory):** Moving beyond coarse-grained experts, MoRAM treats individual rank-1 LoRA adapters as atomic memory units. By shifting from address-based routing to content-addressable retrieval using self-activation mechanisms, MoRAM eliminates explicit routers, preventing routing collapse and significantly improving the plasticity-stability trade-off⁵⁶.

Regularization and Latent Replay

While storing historical data for experience replay provides an empirical upper bound for mitigating forgetting, it is memory-intensive and frequently violates data privacy regulations. Generative replay circumvents this by synthesizing past data; however, in complex generative spaces like diffusion models or LLMs, this often leads to mode collapse.

- **Latent Replay:** Inspired by hippocampal memory consolidation, Latent Replay stores highly compressed feature representations (e.g., hidden states or diffusion latents) rather than raw sensory inputs. In text-to-image diffusion models, retaining and replaying these abstract latent structures dramatically reduces memory overhead while preserving the critical semantic boundaries between sequentially learned visual concepts, effectively preventing rapid utility collapse⁵⁷.
- **TFGN (Task-Free, Replay-Free Architecture):** Operating without replay buffers or task identifiers, the TFGN architecture introduces a structural read/write decomposition at the substrate level of the transformer. The forward pass remains fully dense across all domains, while cross-domain parameter updates are strictly gated to ensure that prior-domain subspaces are never overwritten, realizing autonomous, closed-loop continual learning at the LLM scale⁵⁹.

Methodologies in Model Editing

While continual learning modifies models based on macro-level data distributions, Knowledge-based Model Editing (KME) executes precise, micro-level modifications. The objective is to efficiently correct targeted factual inaccuracies without initiating a retraining cycle and without degrading the model's global performance⁷.

The success of a model editing algorithm is strictly evaluated against four primary criteria⁷:

1. **Reliability (Efficacy):** The ability of the post-edit model to successfully output the target information for the specific edited prompt.
2. **Locality (Specificity):** The preservation of unrelated facts and general linguistic capabilities, ensuring the edit does not bleed destructively into the broader parameter space.
3. **Generality:** The model's capacity to apply the edited fact to paraphrased or structurally divergent queries holding the same semantic intent.
4. **Portability:** The model's ability to logically propagate the edited fact into multi-hop reasoning chains (e.g., if entity A is edited, downstream entity B dependent on A must also reflect the update).

The algorithmic paradigms for model editing generally fall into three methodological families: Locate-then-Edit, Meta-Learning, and Memory/In-Context Editing.

Locate-then-Edit (Direct Parameter Modification)

The locate-then-edit paradigm theorizes that factual knowledge is stored as key-value pairs within the multi-layer perceptron (MLP) modules of specific transformer layers.

- **ROME (Rank-One Model Editing):** ROME utilizes causal mediation analysis to pinpoint the specific mid-layer MLP computations that mediate a factual prediction. Treating the MLP down-projection matrix (W_{down}) as a linear associative memory, ROME computes a precise, rank-one mathematical update to insert a new key-value pair. This enables the overriding of a single fact with exceptional reliability and specificity⁶⁴.
- **MEMIT (Mass-Editing Memory in a Transformer):** Because sequential ROME edits rapidly degrade a model's grammatical cohesion due to gradient interference, MEMIT was developed to scale editing to thousands of facts simultaneously. MEMIT spreads the calculated parameter update across a critical range of multiple MLP layers via a least-squares optimization, effectively buffering the structural integrity of the network⁶⁴. Recent evaluations of MEMIT revealed a vulnerability during "same-subject" batch editing, where identical keys (the subject) are forced to map to divergent values, causing performance collapse. Enhancements like MEMIT-Merge resolve this by unifying the value computation processes for shared subjects⁶⁹.
- **Knowledge Neuronal Ensemble (KNE):** Traditional localization methods suffer from knowledge coupling, where a single neuron corresponds to multiple facts, causing frequent parameter overwriting. KNE identifies distinct ensembles of neurons that encode specific knowledge via gradient attribution scores, applying collaborative error backpropagation exclusively to these targeted ensembles to maintain layer-to-layer

dynamic interaction⁶³.

Meta-Learning and Hypernetworks

Rather than solving a constrained optimization problem for each new fact, meta-learning approaches train an auxiliary neural network to predict the necessary weight changes based on the raw fine-tuning gradient.

- **MEND (Model Editor Networks with Gradient Decomposition):** Fine-tuning an LLM on a single factual correction typically results in catastrophic overfitting. MEND captures the raw fine-tuning gradient and transforms it into a targeted, safe parameter update. To accommodate the massive parameter counts of LLMs, MEND employs a low-rank decomposition of the gradient, rendering it computationally tractable to train a single-hidden-layer MLP hypernetwork that rapidly dictates model edits without damaging unrelated inputs¹⁷.

Memory-Based and In-Context Editing

To entirely bypass the risks of permanent weight degradation, non-parametric editing methods intercept the inference pipeline to alter the model's behavior.

- **SERAC (Semi-Parametric Editing with a Retrieval-Augmented Counterfactual Model):** SERAC stores edits externally in an explicit memory cache. During inference, a scope classifier evaluates the incoming query. If the query aligns with the scope of a cached edit, a smaller, distinct counterfactual model generates the response based on the cached data, leaving the base LLM's frozen weights entirely untouched⁷¹.
- **IKE and ICE (In-Context Editing):** These methods exploit the emergent in-context learning capabilities of LLMs by retrieving updated facts and injecting them directly into the prompt. Consistent In-Context Editing (ICE) formalizes this by establishing a contextual loss function that minimizes the Kullback-Leibler (KL) divergence between the model's output distributions with and without the contextual prompt, gently steering the model to internalize the new knowledge distribution⁷².
- **Dynamic Retriever for IKE (DR-IKE) and FWE-IKE:** To mitigate prompt bloat and redundancy inherent in static k-NN retrieval, DR-IKE employs a reinforcement learning-trained retriever with a dynamic budget controller. This system predicts the hardness of an edit and adaptively prunes low-value demonstrations⁷⁴. Similarly, the Feature-Weighted Ensemble (FWE-IKE) leverages multiple heterogeneous rankers (LLM-, BERT-, and MLP-based) combined with a margin-based routing gate to supply highly calibrated, confidence-aware contextual evidence, drastically improving edit success rates without parameter alterations⁷⁵.

Modality and Linguistic Extensions in Editing

As foundation models expand beyond monolingual text generation, the requirement to edit multimodal and cross-lingual architectures introduces profound structural complexities.

Multimodal Knowledge Editing (MKE)

The transition from text-only LLMs to Vision-Language Models (VLMs) such as LLaVA or CLIP necessitates editing the binding between a visual entity and its textual semantic attributes⁷⁶. A

pervasive, systemic failure mode discovered in VLM editing is "Entity Identity Confusion" (EIC). When an MKE algorithm is employed to alter the identity of a visual subject in an image, the intervention frequently causes severe text-side knowledge contamination. Consequently, text-only queries regarding the original entity hallucinate attributes belonging to the newly edited entity, proving that current MKE techniques fail to maintain strict modality isolation⁷⁸. To circumvent EIC and achieve precise visual interventions, specialized frameworks have been developed:

- **DualEdit:** DualEdit empirically decouples the sensitivity layers of VLMs, discovering that textual and visual representations reach peak sensitivity at distinctly different transformer layers. By introducing a modality-aware gating module, DualEdit applies selective editing adapters to separate layers for visual and textual features, preserving modality boundaries⁷⁹.
- **ReasonEdit:** This framework enhances VLM editing by explicitly incorporating human reasoning chains. When a hallucination is detected, the editor absorbs a chain of factual statements grounded in the image, storing them in an external codebook mapped via a novel topology-balanced multimodal embedding. This reasoning-enhanced retrieval allows the VLM to generalize edits accurately across complex, multi-hop visual scenarios without corrupting isolated text knowledge⁷⁶.
- **LiveEdit:** Operating in lifelong editing scenarios, LiveEdit trains an editing expert generator to produce low-rank experts for individual visual instances. It employs a hard filtering mechanism based on visual semantic knowledge to eliminate irrelevant experts, followed by a soft routing mechanism based on textual relevance to achieve highly stable multi-expert fusion⁸¹.

Cross-Lingual and Multilingual Knowledge Editing

As LLMs are deployed globally, Multilingual Knowledge Editing (MKE) mandates that a factual update applied in a source language (e.g., English) seamlessly propagates to all semantically equivalent queries in target languages (e.g., Chinese, Arabic, German)⁶².

Extensive evaluations utilizing translated benchmarks, such as Multi-CounterFact and Arabic ZsRE, reveal that existing locate-then-edit methods (e.g., ROME, MEMIT) fail catastrophically in cross-lingual settings⁸³. This failure is attributed to *language anisotropy*: multilingual foundation models encode different languages into misaligned, segregated latent subspaces. A parameter update targeting an English subject's key-value representation physically cannot intersect with the corresponding Chinese or Arabic representation⁸³. Research indicates an asymmetry in transfer, wherein edits injected via English rarely propagate to German, and Arabic edits fail to influence English queries⁸⁴.

Overcoming language anisotropy requires explicit cross-lingual structural alignment. Frameworks such as CLICKER address this by avoiding offline parameter updates. Instead, CLICKER utilizes relevance-aware dynamic retrieval and stepwise in-context reasoning to enforce language alignment dynamically during the generative process, demonstrating vastly superior cross-lingual consistency compared to static editing baselines⁸⁵.

Challenges and Prospects

While continual learning and model editing have made significant strides, the deployment of these technologies in highly scaled foundation models reveals persistent vulnerabilities. The following subsections analyze the most pressing open challenges: multi-hop ripple effects, sequential degradation, the nuanced boundary between editing and machine unlearning, and security implications.

The Ripple Effect and Logical Consistency

The most critical limitation of contemporary knowledge editing is the "ripple effect"⁸⁸. A robust knowledge editing system must not only update the specific targeted fact (e.g., replacing the CEO of a company) but must also logically propagate this edit to all dependent, multi-hop queries (e.g., answering questions regarding the CEO's spouse, corporate headquarters, or historical timeline based on the new fact)⁸⁸.

Diagnostic benchmarks such as MQuAKE and RippleEdits systematically test these multi-hop dependencies⁶. Evaluations reveal a stark contrast in performance: while parameter-modifying methods like MEMIT achieve >90% success on direct queries, their logical generalization and multi-hop ripple success rates frequently plunge below 20%⁸⁹. The underlying cause of this failure has been quantitatively linked to GradSim—the cosine similarity between the gradients of the original fact and its logically related knowledge. If logically related concepts are stored in disparate, orthogonal parameter spaces (low GradSim), a localized edit to one fact is mathematically incapable of rippling to the other, leading to profound logical inconsistencies⁸⁹. To enforce multi-hop consistency, advanced interventions explicitly map the broader semantic neighborhood.

- **K-Edit:** This framework incorporates external Knowledge Graphs (KGs) during the editing process. Because KGs maintain explicit relational edges, editing a single node automatically generates contextual secondary edits, which are then batch-applied to the language model, significantly boosting multi-hop reasoning¹⁵.
- **EchoEdit:** EchoEdit tackles the ripple effect by constructing a RippleGraph to model the potentially affected knowledge regions explicitly. It utilizes a dynamic RippleRule generator to derive specific diffusion rules that precisely constrain knowledge propagation, preventing both under-rippling (failure to update) and over-rippling (erroneously altering unrelated facts)⁹³.
- **IRAKE:** For retrieval-augmented editors, the "edit skipping" phenomenon—where the model ignores retrieved facts during multi-hop reasoning—is mitigated by Iterative Retrieval-Augmented Knowledge Editing (IRAKE). This method guides the LLM to decompose complex questions recursively, utilizing a state backtracking mechanism to ensure the edited fact influences the final deductive output⁹⁴.

Scalability and Sequential Degradation

While batch editing algorithms like MEMIT can process up to 10,000 facts simultaneously under experimental conditions, real-world deployment requires *sequential* lifelong editing, where

solitary edits stream in continuously over extended periods. Research conclusively establishes that sequential parameter editing induces "knowledge attenuation" and gradual catastrophic forgetting⁹⁵.

As the number of sequential edits increases, the cumulative effect of low-rank perturbations or rank-one matrix updates exhausts the structural capacity of the transformer's feed-forward layers. Previously edited facts are aggressively overwritten by newer edits, and the model's baseline grammatical and reasoning capabilities degrade significantly⁹⁵. Transitioning away from direct weight modification toward memory-based routers, in-context learning, or dynamic MoE architectures (e.g., TIME, which isolates sequential updates using tensor-factorized intrinsic experts) offers the most viable pathway to circumvent parameter saturation in lifelong editing scenarios⁹⁷.

The Intersection of Model Editing and Machine Unlearning

As foundation models face increasing scrutiny under global data protection regulations (e.g., GDPR, CCPA), the concept of Machine Unlearning has gained immense traction. In its strictest classical formulation, machine unlearning is a dataset-defined deletion problem: given a

specific "forget set" ($D \setminus F$), the objective is to produce a model whose behavior and latent parameters are mathematically indistinguishable from a counterfactual model trained entirely without that data⁹⁸.

In the realm of foundation models, exact retraining-based unlearning is computationally impossible. Consequently, the field has gravitated toward approximate unlearning and behavioral suppression. Techniques such as Representation Misdirection Unlearning (RMU) explicitly disrupt the intermediate latent representations of sensitive data, steering the model's activations toward a predetermined, random vector whenever hazardous or copyrighted data is queried⁹⁹.

Increasingly, researchers are reformulating machine unlearning as a specialized application of Model Editing¹⁸. In this paradigm, rather than injecting a new factual association, algorithms like ROME, MEMIT, or WISE are employed to remap sensitive input queries to a targeted "avoidant" or "refusal" vector (e.g., an <EOS> token or a standard "I cannot answer this" response)¹⁰⁰.

Extensive evaluations on synthetic benchmarks like TOFU (Task of Fictitious Unlearning) and hazardous knowledge benchmarks like WMDP reveal a complex trade-off¹⁰¹. Model editing methods can achieve an exceptionally precise "Forget Quality," often outperforming traditional gradient ascent or divergence minimization unlearning baselines in erasing specific instances. However, because editing methods strictly optimize highly localized mappings, they frequently fail to encapsulate the broader semantic scope of the unlearned concept—a phenomenon termed "under-spreading." While an edit may successfully induce a refusal for an exact string query, adversarial paraphrases or multi-hop logical deductions may bypass the localized edit and extract the "forgotten" knowledge. Conversely, when edits are aggressively scaled to enforce broad forgetting, they rapidly induce catastrophic structural damage to the LLM's general capabilities (consistency collapse)¹⁰¹. Thus, utilizing model editing for true machine unlearning remains highly effective for targeted surgical strikes but proves brittle against

generalized adversarial extraction.

Security Implications and Malicious Misuse

The extreme efficiency of In-Context Editing (IKE) and direct parameter modification introduces profound security vulnerabilities. Malicious actors can exploit these techniques to silently inject misinformation, systemic bias, or toxic behaviors into open-source models or via high-level wrapped APIs where the final prompt is obscured from the end-user.

Detection benchmarks, such as KETIBench, evaluate the capability of models to identify manipulated knowledge. Studies demonstrate that while simple classifiers can detect the presence of an in-context edit with high accuracy by analyzing the entropy and ranking of next-token output probabilities, distinguishing between a benign factual correction and a malicious manipulation remains exceedingly difficult¹⁰⁴. The development of robust "reversal tokens" and active prompt-guarding mechanisms to detect, quarantine, and neutralize unauthorized interventions represents a critical, unresolved frontier for AI safety and trustworthiness¹⁰⁵.

Conclusion

The transition from static, monolithically trained foundation models to dynamic, continuously evolving intelligent systems represents one of the most critical trajectories in contemporary artificial intelligence research. Continual Learning and Model Editing offer complementary methodological frameworks for achieving this dynamic adaptability. Continual learning approaches—empowered by advances in parameter-efficient fine-tuning, optimal policy regularizers, dynamic prompting, and latent replay—provide robust mechanisms for broad domain adaptation and alignment tracking while mitigating the historically intractable problem of catastrophic forgetting. In parallel, model editing techniques, ranging from precision rank-one weight modifications to dynamic in-context retrieval and hypernetworks, grant practitioners the surgical ability to rectify hallucinations and update discrete factual associations in real time.

Despite rapid and profound advancements, severe challenges remain. The inability of current editing methods to reliably navigate the multi-hop ripple effects of interconnected knowledge, the deep language anisotropy hindering cross-lingual propagation, and the modality contamination observed in vision-language models underscore the extreme geometric complexity of latent representations. Furthermore, bridging the theoretical gap between localized behavioral editing and verifiable, mathematically rigorous machine unlearning is imperative for compliance with global data privacy and safety regulations. Future progress relies on synthesizing these paradigms: developing novel architectures that intrinsically support modular, interference-free continuous updates, while embedding deep semantic logic that guarantees factual consistency across the model's entire reasoning space. By resolving the fundamental tension between plasticity and stability, the AI community moves closer to deploying lifelong foundation models that are simultaneously resilient, hyper-accurate, and perfectly aligned with a continuously evolving world.

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